[illegible]

[illegible]

[illegible] A [illegible] Tick [illegible]500ms [illegible]A/B/C/D/E[illegible] Tick [illegible] Tick [illegible] E [illegible]Return5min[illegible]

[illegible] Kyle[illegible]1985[illegible] Glosten-Milgrom[illegible]1985[illegible] 33 [illegible] 59 [illegible]TI[illegible]OVI[illegible]OBI[illegible] 28 [illegible] Ridge [illegible]$\alpha \in [10, 1000]$[illegible] EWMA [illegible] IC [illegible] Softmax [illegible]

[illegible] Leave-One-Model-Out[illegible]LOO[illegible] 54 [illegible] 28[illegible]IC [illegible] 0.313 [illegible] 0.330[illegible] ICIR [illegible] 7.82 [illegible] 12.59[illegible]+61%[illegible]-Niche [illegible]e_sect_lag_gap[illegible]IC=0.203[illegible] [illegible]LODO-CV[illegible] 20 [illegible] CV [illegible] IC=0.3302[illegible]ICIR=12.59[illegible] [illegible]IC=0.147[illegible] 124.6%[illegible]+0.051[illegible] [illegible] O(1)[illegible] Alpha [illegible]

[illegible]Ridge [illegible] IC [illegible]Leave-One-Day-Out [illegible]

[illegible]

1.1 [illegible]

[illegible]A/B/C/D/E[illegible] 5 [illegible] 27,601 [illegible] Tick [illegible] 33 [illegible]

[illegible] E [illegible]

$$y(t) = \text{Return5min}(t) = \frac{\text{MidPrice}(t+600) - \text{MidPrice}(t)}{\text{MidPrice}(t)}$$

[illegible] $\text{MidPrice}(t) = \left(p_1^{b(t)} + p_1^{a(t)}\right)/2$[illegible]600 [illegible] Tick [illegible] 5 [illegible]

[illegible] $\hat{y}(t) = f_\theta(\boldsymbol{x}_1, ..., \boldsymbol{x}_t)$ [illegible] t [illegible] t [illegible]

[illegible]

- [illegible]IC[illegible]$\text{IC}_d = \text{Pearson}(\hat{\boldsymbol{y}}_d, \boldsymbol{y}_d)$[illegible][1][illegible] 53 [illegible]
- ICIR[illegible]$\text{ICIR} = \overline{\text{IC}}/\sigma(\text{IC})$[illegible]IC [illegible][1][illegible] 55 [illegible]

1.2 [illegible]

[illegible]Market Microstructure Theory[illegible][2][illegible]

1.2.1 Kyle[illegible]1985[illegible]

Kyle[illegible]1985[illegible][3] [illegible] u_t[illegible]

$$\Delta m(t) = \lambda \cdot u_t + \varepsilon_t$$

[illegible] $\lambda > 0$ [illegible] Kyle's Lambda[illegible]ε_t [illegible]TI[illegible]OVI[illegible]——[illegible]Cont [illegible]2014[illegible][4] [illegible]OFI[illegible]

1.2.2 Glosten-Milgrom[illegible]1985[illegible]

Glosten-Milgrom[illegible]1985[illegible][5] [illegible]——[illegible] OB[illegible]2–5 [illegible]"[illegible]"[illegible]

1.2.3 [illegible]AMH[illegible]

Lo[illegible]2004[illegible][6] [illegible]Adaptive Markets Hypothesis[illegible]AMH[illegible]AMH[illegible]Day2[illegible] Day4[illegible]

1.3 [illegible]

1.3.1 [illegible]OVI[illegible]

[illegible]OVI[illegible] IC≈0.26[illegible]

$$\mathrm{OVI}(t) = \frac{\sum_{i=1}^{5} V_i^{b(t)} - \sum_{i=1}^{5} V_i^{a(t)}}{\sum_{i=1}^{5} V_i^{b(t)} + \sum_{i=1}^{5} V_i^{a(t)} + \varepsilon} \in [-1, 1]$$

□□ $\varepsilon > 0$ □□□□□□□□□□□□V_i^b, V_i^a □□□□ i □□□□□□□□□□

“OVI □□”□Pulse□□□□□□ OVI □□□□□ SMA □□□□□□□□□□□□□

$$\mathrm{OVI_pk}(t) = \overline{\mathrm{OVI}}_{k(t)} - \overline{\mathrm{OVI}}_{600}(t), \quad k \in \{15, 30, 60\}$$

□□ $\overline{\mathrm{OVI}}_{k(t)} = \frac{1}{k} \sum_{s=t-k+1}^{t} \mathrm{OVI}(s)$ □□□ k □ Tick □□□□□□□□□ $k < 600$ □□OVI_pk > 0 □□□□□□□□□□□□□□□□□□□□□□□□

1.3.2 □□□□□□TI□

□□□□□□□□□

$$\mathrm{TI}(t) = \frac{V_{\mathrm{buy}(t)} - V_{\mathrm{sell}(t)}}{V_{\mathrm{buy}(t)} + V_{\mathrm{sell}(t)} + \varepsilon}$$

□□ $V_{\mathrm{buy}}, V_{\mathrm{sell}}$ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□TI □□□□ Kyle □□□□□□□□ u_t□□□□□□□□□□□□□

1.3.3 □□□□□□□□OBI□

□□ Glosten-Milgrom □□□□□□□□□2–5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□

$$\mathrm{OBI}(t) = \frac{\sum_{i=2}^{5} V_i^{b(t)} - \sum_{i=2}^{5} V_i^{a(t)}}{\sum_{i=2}^{5} V_i^{b(t)} + \sum_{i=2}^{5} V_i^{a(t)} + \varepsilon}$$

□□ OVI□1–5 □□□□□OBI □□□□“□□□□”□□□□□□□□□□ OVI □□□□□□□□□□□

1.4 □□□□

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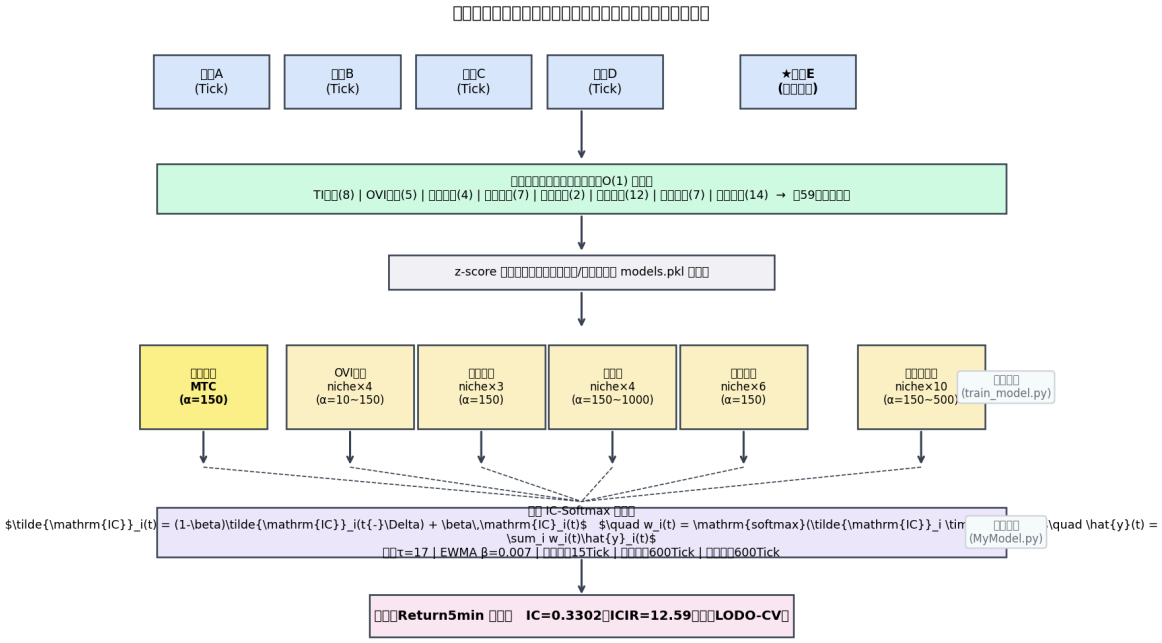

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□□□□□□□□□`train_model.py`□□□□□□□□□□□□□□□□

[illegible]MTC[illegible]1 [illegible] 2 [illegible]27 [illegible] Niche [illegible] 0[illegible] IC [illegible] IC [illegible]

1.6 [illegible]

1.6.1 LOO [illegible]

[illegible]“[illegible]”[illegible]5 [illegible] Leave-One-Model-Out [illegible]LOO[illegible] $\mathrm{OM} = \overline{\mathrm{IC}} - 0.5\sigma(\mathrm{IC})$[illegible]LOO [illegible]

$$\Delta\mathrm{OM}_m = \mathrm{OM}(\square\square\square) - \mathrm{OM}(\square\square\ m)$$

[illegible] $\Delta\mathrm{OM}_m < 0$[illegible] m [illegible][7][illegible]54 [illegible] 28 [illegible]ICIR [illegible] 7.82 [illegible] 12.59[illegible]+61%[illegible]

1.6.2 [illegible]-Niche [illegible]

[illegible]“[illegible]”[illegible]“Niche [illegible]”[illegible]“[illegible]”[illegible]“[illegible]”[illegible] Day1 [illegible]

1.6.3 [illegible]

[illegible] 7 [illegible]“[illegible] OVI”[illegible]e_sect_lag_gap[illegible]

$$\mathrm{e_sect_lag_gap}(t) = \mathrm{OVI_p15}^{E(t)} - \mathrm{Sect_OVI_p20}(t)$$

[illegible] E [illegible] OVI [illegible] E [illegible]/[illegible] IC [illegible] 0.203[illegible]

1.6.4 [illegible]

[illegible] 59 [illegible] $O(1)$ [illegible]SMA [illegible]EMA [illegible][8][illegible]

[illegible]

2.1 [illegible]

2.1.1 [illegible]

[illegible] E [illegible]

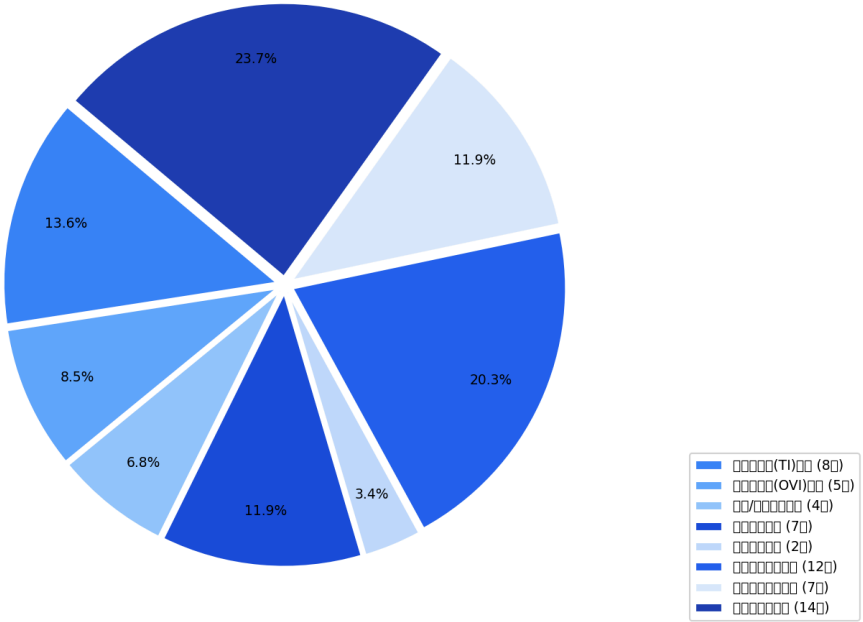
$$y(t) = \frac{\mathrm{MidPrice}(t+600) - \mathrm{MidPrice}(t)}{\mathrm{MidPrice}(t)}$$

$t + 600$ [illegible] t [illegible] t [illegible] $\hat{y}(t)$[illegible] $y(t)$ [illegible] IC [illegible]

2.1.2 [illegible]

[illegible](1) [illegible]——[illegible] t [illegible](2) [illegible] IC [illegible]——[illegible] 5 [illegible] IC [illegible](3) [illegible]——[illegible] < 0.7[illegible](4) [illegible]——[illegible]

2.1.3 [illegible]59 [illegible]8 [illegible]



□ 2 □□□□□□□□□□□ 59 □□□□8 □□□

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(1) □□□□□□TI□□□□8 □□□ TI_p$k = \overline{\mathrm{TI}}_k - \overline{\mathrm{TI}}_{600}$□$k \in \{15, 30, 40, 60\}$□□□□□□□□□□□□□□□□□□□□□ EMA □□ TI_ep$k$□

(2) □□□□□□OVI□□□□5 □□□ □□ 1.3.1 □□□□□□ IC≈0.26□□□□□□□□

(3) □□/□□□□□□□4 □□□ □□□□□□□□□□ ON□□Order Number Imbalance□□ TN□□Trade Number Imbalance□□□ OVI □□□□□□□□□□□□□

(4) □□□□□□□7 □□□ A/B/C/D □□□□□□□□□□□□□□□□□ OBI□□□ TI □□□□□ OVI □□□□□□□□□□□□□□□□□□

(5) □□□□□□□2 □□□ `aft_13800`□□□□□□□$t \geq 13800$s□□ `aft_12000`□□□□□□□□□□□□□

(6) □□□□□□□□□12 □□□□□ E □□□□□□ 5 □□□□□□□□□□□□□□□□□□□ E □ 30/60/120/300/600/900 Tick □□□□□□□

(7) □□□□□□□□□7 □□□ □□□□□□□□□OBI□2−5 □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□“□□□□”□□□□□□

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2.2 □□□□□

2.2.1 □□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□ $O(1)$ □ Tick□

- `_RunSMA(window)`□□□□□□□□□□□□□□□□□□□deque□□□□□□
- `_RunEMA(span)`□□□□□□□□□$\mathrm{EMA}_t = (1-\alpha) \cdot \mathrm{EMA}_{\{t-1\}} + \alpha \cdot x_t$□$\alpha = \frac{2}{\mathrm{span}+1}$□

2.2.2 [illegible]

[illegible] k [illegible] Tick[illegible]k [illegible] 0 [illegible] 0 [illegible][9][illegible]

2.2.3 [illegible]

Ridge [illegible] z-score [illegible]$\hat{x} = (x - \mu)/\sigma$[illegible] μ, σ [illegible] `models.pkl`[illegible]

2.2.4 [illegible]

[illegible]

- [illegible]$\geq$ 120 Tick[illegible]$x \leftarrow \mathrm{clip}(x, -0.1, 0.1)$
- [illegible]30/60 Tick[illegible]$x \leftarrow \mathrm{clip}(x, -0.05, 0.05)$
- [illegible] $[-0.5, 0.5]$[illegible]

2.3 [illegible]

2.3.1 [illegible]**LODO-CV**[illegible]

[illegible] 5 [illegible]Fold [illegible] Leave-One-Day-Out [illegible]LODO-CV[illegible][10][illegible] 5 [illegible] 4 [illegible]IC [illegible] IC [illegible] `GroupKFold(n_splits=5)` [illegible] `Day` [illegible]

2.3.2 [illegible]

[illegible]Nested Blind Test[illegible]

- [illegible] IC[illegible] [illegible] 4 [illegible]
- [illegible] IC[illegible] [illegible] 4 [illegible] 4 [illegible] CV[illegible] 4 [illegible]
- [illegible]-[illegible] gap[illegible] Iter20 [illegible] gap [illegible] -0.019[illegible]

2.4 [illegible]

2.4.1 Ridge [illegible]

[illegible] L_2 [illegible]Ridge Regression[illegible][11][illegible]

$$\min_{\boldsymbol{w}, b} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{w} - b\|_2^2 + \alpha \, \|\boldsymbol{w}\|_2^2$$

[illegible]$\hat{\boldsymbol{w}} = \left(\boldsymbol{X}^\top \boldsymbol{X} + \alpha \boldsymbol{I}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}$

Ridge [illegible] OLS [illegible](1) [illegible] OVI/TI [illegible]Ridge [illegible] α [illegible]-[illegible](2) [illegible] LightGBM/[illegible]Ridge [illegible] $n = 138000$[illegible] $p = 59$ [illegible] α [illegible] $\alpha = 150$[illegible]OVI [illegible] $\alpha = 10$[illegible] $\alpha = 1000$[illegible][11][illegible]

2.4.2 28 [illegible]

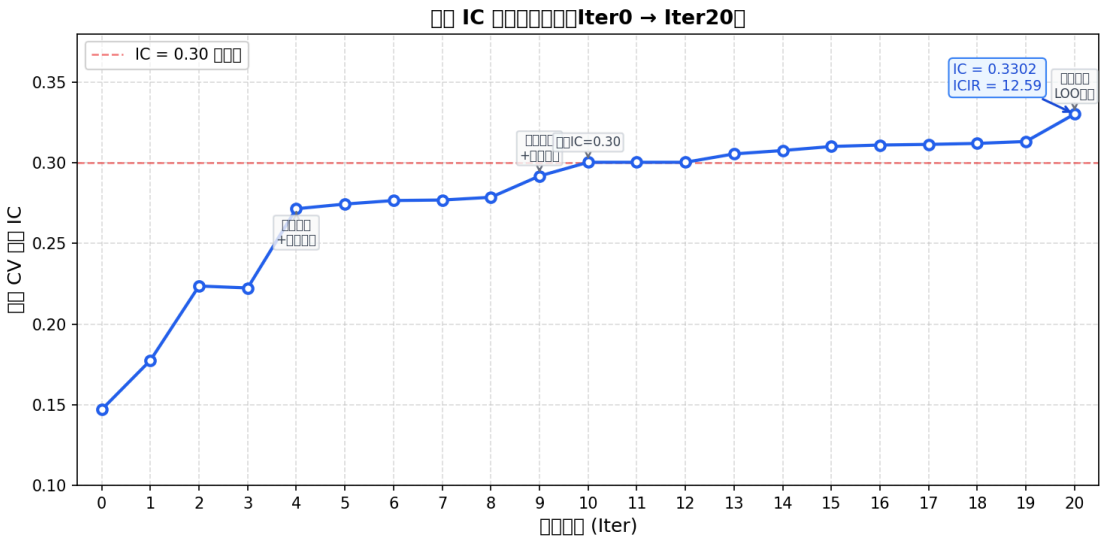
28 [illegible]1 [illegible]MTC[illegible]27 [illegible] Niche [illegible]

□□□□	□□	□□□ α	□□□□
□□□Stable□	1	150	TI □□ + □□□□□MTC□
OVI □□ niche	4	10–150	OVI □□□□□□□□□
□□□□ niche	3	150	`sect_ret_lag` + `e_ret_lag` + `e_sect_lag_gap`
□□□ niche	4	150–1000	`spread_wt_ovi`□`ovi_sq`□□□□□
□□/□□ niche	6	150	`obi_deep_p15`□`cum_flow_imb`□`book_pres_pulse`
□□□□□ niche	10	150–500	□□□□□

□ 1　□□□□□□□□

2.4.3　□□□□□□

□□ 2 □□□□□□□□ 20 □□□□□□□IC □ 0.147 □□□□□ 0.3302□



□ 3　□□ IC □□□□□□□□Iter0⟶Iter20□

□□□□□□□Iter0–2 □□ LODO-CV □□□□□IC □ 0.147 □□□ 0.224□Iter3–8 □□□□□□□□□IC □□□ 0.279□Iter9–12 □□□□□□□□□□□□□IC □□□□ 0.30□Iter13–19 □□□□□□□□□□□□□□□□□ OVI □□□□□□IC □□□□□ 0.313□ICIR □ 7.82□Iter20 □□ LOO □□□54 □□□□□□ 28 □□IC □□□ 0.330□ICIR □ 7.82 □□□ 12.59□+61%□□

2.4.4　□□□□□□

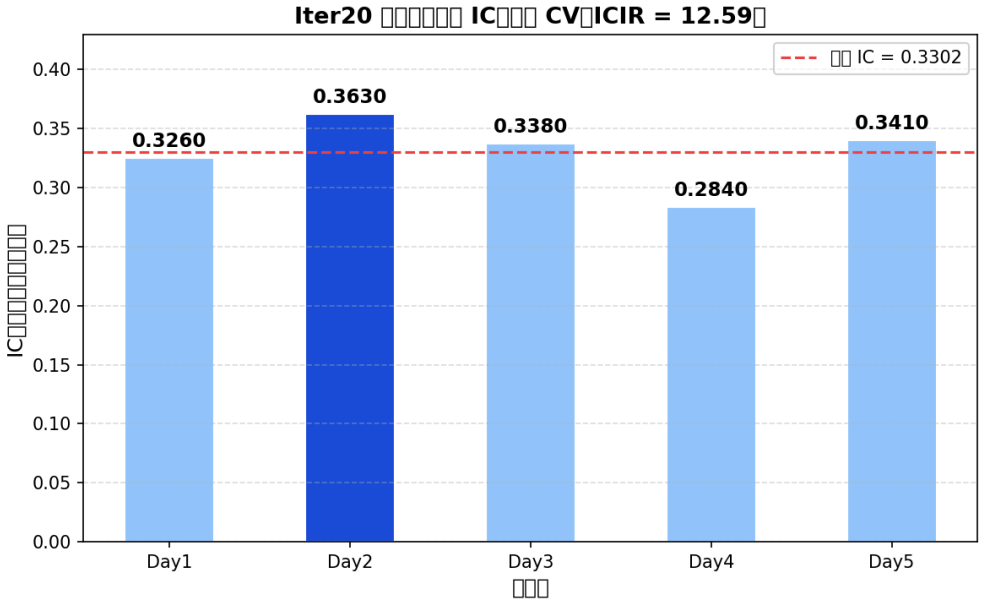
□□	□	□□
`ENSEMBLE_WINDOW`	600	□□ IC □□□□□□ 5 □□□
`ENSEMBLE_TEMP`	17	Softmax □□□□□□□□□□
`ENSEMBLE_FLOOR`	0	□□□□□□□□□□ 0□
`RETURN_DELAY`	600	IC □□□□□□□□Tick□
`ENSEMBLE_UPDATE_FREQ`	15	□□□□□□□ 15 Tick□
`ENSEMBLE_EWMA_BETA`	0.007	IC □□□□□□□□ 99 Tick□
`STABLE_PRIOR`	2.0	□□□□□□□□□□□□□
`NICHE_INIT_WEIGHT`	0	Niche □□□□□□□□□

□ 2　□□□□□□□□

2.5 □□□□□□□□□□□

2.5.1 □□ IC □□

□ 3 □□□□□□□□□Iter20□□□□ CV □□ IC□

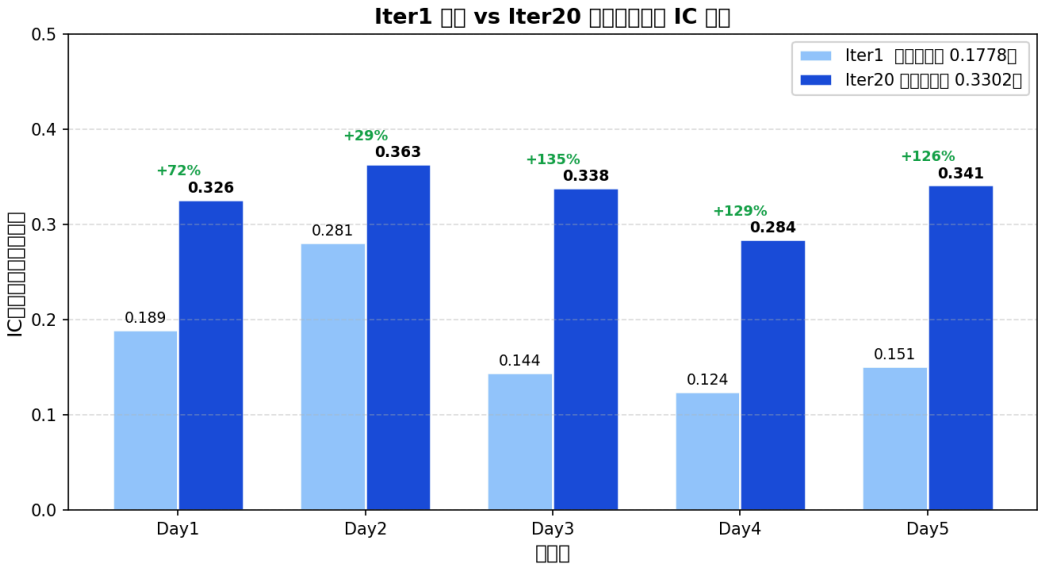


□ 4　Iter20 □□□□□□ CV □□ IC□□□=0.330□ICIR=12.59□

□□□□ IC □□□□Day1=0.326□Day2=0.363□Day3=0.338□Day4=0.284□Day5=0.341□□□ IC=0.3302□□□□=0.026□ICIR=12.59□Day4□□□□□□□IC □□□0.284□□□ AMH □□□□——□□□□□□□□□□□□□□□□□□□□□□□□□□□□ IC □□□□ 0.026□□□□□□□□□□□□□□

2.5.2 □□□□

□ 4 □□□□ Iter1□6 □□□□□□□□□ Iter20□28 □□□□□□□□□□□ IC □□□

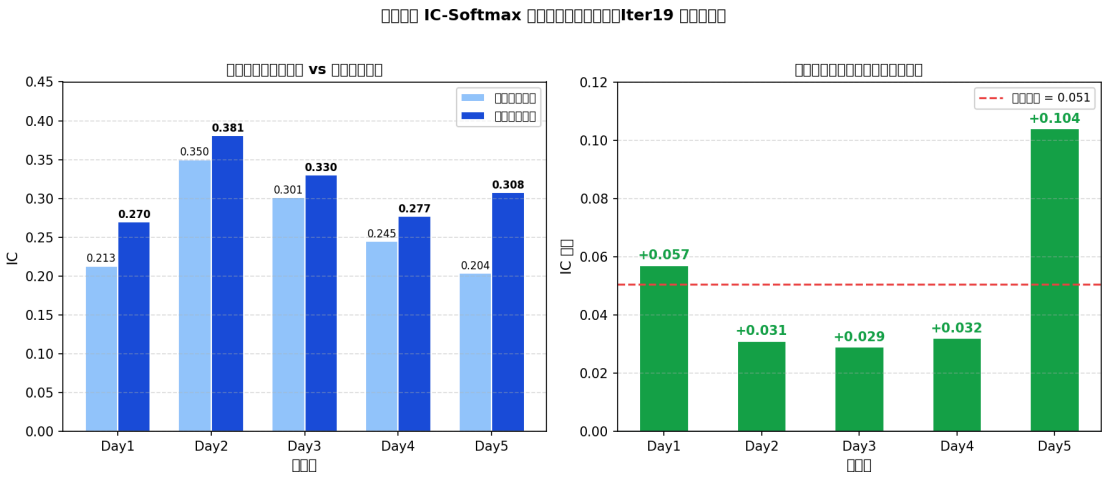


□ 5　Iter1 □□ vs Iter20 □□□□□□ IC □□

□□□□□Iter20 □□□□□□□□□□□□□□Day1 □□ 72.5%□0.189⟶0.326□□Day3 □□ 134.7%□0.144⟶0.338□□Day4 □□ 129.0%□0.124⟶0.284□——□□□□□Day1/3/4□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

2.5.3 □□□□□□□□

□ 5 □□□□□□□□□□□□□□□□□□□□□□□□□□



□ 6　□□□□ vs □□□□□□□ IC □□□□□□□□

□□□□□□□ 5 □□□□□□□□□□□□□□+0.051□□□□□□ IC-Softmax □□□□□□[12]□□□□□□□ Day5□+0.104□□□□ Niche □□□□□ IC □□□□□□□□□□□□□□□□□□□□□

2.5.4 □□□□□□

□□□□	□□□	□□
□□ CV □□ IC	0.3302	□□□□□□□out-of-sample □□
□□ CV IC □□□	0.026	□□□□□□□□□□
ICIR□IC/Std□	12.59	□□□□□ IC□□□□□
Penalized□M—0.5S□	0.317	□□□□□□□□□□□□□
□□ 4 □ IC	0.311	□□□□□□□□□□□□□□
□-□ gap	—0.019	□□=□□□□□□□□□□□□
□□□□□vs □□□	+0.051	□□□□□□□□□□□□
□□□□□	28 □□□□59 □□	□□ Iter19□54×59□□□□

□ 3　□□□□□Iter20□□□□□□□

□□□□□□□□□□□□□□□□□□□□□(1) □□□ $\hat{y}(t)$ □□□□□□□□□□□□□□□□□□□□

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